Nishant Bhave

Final Year, B.Tech + M.Tech
Electrical Engineering and Controls
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University	Degree	Year	GPA
IIT Bombay	B.Tech Electrical Engineering, M.Tech Controls and Computing	2022–27	9.12

Pursuing a **Dual Minor** in **Robotics** and **Machine Learning (C-MInDS)**

Scholastic Achievements

- Received **Undergraduate Research Award** for bilevel optimization research with Dr. Baranwal [2025]
- Secured **All India Rank 964** in the JEE Advanced out of **0.16 million+** candidates [2022]
- Qualified for **scholarship** and acquired **AIR 262** in Kishore Vaigyanik Protsahan Yojana [2022]
- Received **Silver Medal** in Dr. Homi Bhabha Balvaidnyanik Science Talent Search Competition [2020]

Master Thesis

ANTM: Assertive Negotiation Traffic Model [Apr'26 - Present]
Guide: Prof. Parthib Khound, IIT Bombay and Honda R&D

- Developed **ANTM**, a microscopic traffic model reproducing dense lane-less traffic for **development and testing** of autonomous-vehicle planning and control
- **Negotiation by assertion**: a vehicle claims contested space by physically entering it and infers concession from a **counterfactual test**, giving the follower vehicle a principled way to refuse
- **Density-adaptive resolution**: line spacing selected from the local **packing fraction** over nested line sets with hysteresis, so free-flowing and jammed traffic still share one coordinate system
- Evaluated on a controlled three-vehicle bench and open-road simulation under a **graded contact-severity** metric suite (severe-contact and brush rates, TTC exposure, fundamental diagram, pass clearance)

Situation-Aware Feedback-Predictive Control *Published at CoDIT 2026* [Jan'26 - Jun'26]
Guide: Prof. Parthib Khound, IIT Bombay and Honda R&D

- Developed a **hybrid feedback-predictive** framework for navigation in dense unstructured traffic on **kinematic bicycle model**, assuming no lane structure or behavioural compliance from neighbours
- Combined **360° zone-based perception** with a dual-layer strategy: longitudinal **PD feedback** on a reference speed set by braking distance and steering dynamics, and lateral tracking of a **virtual reference lane** computed from the spatial distribution of neighbours
- **Predictive planner** samples control inputs over a horizon and selects the most feasible trajectory under a **multi-term cost**, validated across diverse one-way dense-traffic scenarios

Research Experience

Fixed-Work Decentralized Bilevel Optimization *Under Review, AISTATS* [May'26 - Present]
Guide: Prof. Mayank Baranwal, IIT Bombay

- Proposed a **single-outer-loop** decentralized bilevel method performing one **Nesterov extrapolation**, a fixed number of **Chebyshev-accelerated gradient-tracking** correction sweeps for the lower and **adjoint** states, and one tracked outer update per iteration
- Correction length is **independent of the target accuracy** ϵ, so that no accuracy-dependent nested solve, exact lower-level solution, **inverse Hessian** or exact hypergradient is ever required
- Proved $\tilde{\mathcal{O}}(\sqrt{\chi}\sqrt{\kappa_x}P_y\log(1/\epsilon))$ **communication complexity**, retaining accelerated $\sqrt{\chi}\sqrt{\kappa_x}$ dependence of the single-level benchmark and established **two lower bounds**, on communication rounds and oracle calls

Reach-Avoid MPC with Guaranteed Recursive Feasibility *Under Review* [Oct'25 - Dec'25]
Guide: Prof. Shankar Deka, Aalto University

- Propagated **actuator limits** through the **ECGBF** backstepping construction to synthesize an **input-constrained reach-avoid invariant set**, with input bounds as linear constraints in the controller
- Used that set as the MPC **terminal set** to prove **recursive feasibility**, guaranteeing an admissible input at every subsequent step while ensuring safety and reachability into the target set
- Showed the **zero-order-hold** implementation preserves the continuous-time guarantees under sufficiently **fast sampling**, yielding a sampled-data MPC framework for control-affine **nonlinear** systems

Transferable Neural CBFs via Generalized Closure *Under Review at ACC* [Mar'25 - Aug'25]
Stochastic Robotics Lab, Prof. Shishir Kolathaya, Indian Institute of Science Bangalore

- Proved a **generalized closure condition** and **lifting theorem**: a barrier condition verified on a compact set of per-obstacle **relative coordinates** holds for every obstacle in every map, with the information those coordinates discard made explicit as a bounded **residual**
- Derived a **rank condition**, checkable before training, that is necessary for symmetry to guarantee closure, and gave explicit residual sets for three systems
- Identified **body-frame** coordinates that close exactly for a full **12-state quadrotor**, deploying the learned barrier on **unseen maps without retraining**

Gradient-Flow Methods for Bilevel Optimization *Accepted at ICC 2027* [Jul'25 - Dec '25]
Guide: Prof. Mayank Baranwal, IIT Bombay

- Cast bilevel optimization as a **continuous-time gradient flow** driven by a hypergradient estimator and analyzed it with **Lyapunov stability theory** across three distinct regularity regimes
- Utilized a **normalized gradient flow** reaching the **global optimum in finite time** when both levels are strongly convex, established via $\dot{V} \leq -k\sqrt{V}$
- Relaxed to **Polyak-Łojasiewicz** conditions: a PL upper level converges to a residual neighbourhood sized by problem **conditioning**, and a **non-convex** upper level converges asymptotically under the standard flow and rigorous **hypergradient error bounds**

SPART: 2-D LiDAR Emulation for AV Testing [May'24 - Mar'25]
Guide: Prof. Leena Vachhani, Centre for Systems & Control, IIT Bombay

- Developed an open-source tool synthesizing **geometrically exact** 2-D LiDAR point clouds directly from real-world vehicle **trajectory datasets**, requiring no graphical simulator, GPU or manual scene design
- Combined **range-gated temporal eligibility** scheduler, **angular interval filter** and **Numba JIT**-compiled scan kernel to cut ray-segment intersection tests **60**× relative to brute-force ray casting
- Validated on ≈**35,000** frames across four **INTERACTION** dataset sessions at **1,648 Hz** full-pipeline throughput, **164**× real time, with **zero** false-positive hits and **zero** range error

Vision-based Unmanned Aerial Manipulator Control [May'25 - Jul'25]
Guide: Prof. Lin Zhao, CORE Lab, National University of Singapore

- Evaluated pose estimation frameworks and deployed **FoundationPose** for real-time 6D object tracking
- Utilized hybrid **MPC + L1 adaptive controller** for trajectory tracking under disturbances during dynamic high-speed pick-up maneuvers, accounting for uncertainties in model parameters.
- Demonstrated vision-based closed-loop grasping of a can in both PX4-Gazebo simulation and **real-world drone experiments** with **YOLO**-based **online segmentation** for initial mask generation.

Professional Experience

Robotics R&D Intern | ABB Innovation Centre | Robotics Lab [May'24 - Jul'24]

- Researched, compared, and implemented an optimal two-phase Recursive Partitioning Algorithm in `.NET C#` for single-SKU palletizing, a variant of **2D Bin Packing** problem
- Developed a mixed SKU pattern generator, **single-handedly** engineering an **algorithm** with 4 variants to ensure gripper safety margins, account for placement errors, and **enhance stability** during transportation
- Implemented the **recursive algorithm** with **backtracking** for feasibility checks and robust performance

Key Projects

Fault-Tolerant Controller for Single Rotor Failure | Inter-IIT Tech Meet [Oct'24 - Dec'24]

- Secured **2nd** position for developing a fault-tolerant controller for a quadrotor under **single-rotor failure**
- Employed **NDI** combined with **feedback linearization** for quick response to motor failure and stabilization
- Implemented a modified **Two-Stage Kalman Filter** for real-time failure detection using only the onboard IMU and GPS. It generates a control-effectiveness matrix, which adapts to motor failure within **0.03s**.
- **Validated** control and fault-detection algorithms in Gazebo simulation as well as **hardware tests**, relying solely on the PX4 flight controller stack, with no external computation devices or additional sensors.

Thanopod | Team genSUS, IIT Bombay | Robofest Gujarat 4.0, GUJCOST [Jun'24 - Jan'25]

- Conducted a **literature review** on the control of self-balancing bots and autonomous navigation
- Simulated motor dynamics for **Thanopod 2-Wheeled Self-Balancing Bot** to model the non-linear system for designing **LQR** Controller for enhanced stability, estimation and precise motion control
- Designed a **cascaded PID** loop ensuring stable balance with smooth motion along a **curvilinear** track
- Applied advanced **computer vision** and **stanley controller** for effective **line detection** and **tracking**

Adversarial Training for Ergodic Reward [Oct'25 - Mar'26]

Guide: Prof. Dominik Baumann, Cyber-Physical Systems Group, Aalto University

- Used a **Wasserstein GAN** to learn a reward transformation rendering **non-ergodic** environments ergodic, so that standard RL algorithms maximizing **expected** reward apply unchanged
- A generator transforms the **time-average** rewards while a discriminator evaluates them against **ensemble-average** rewards, with accuracy ≈ 0.5 indicating the two have become indistinguishable

Teaching Assistantships

- **Distributed Optimisation and Machine Learning** | Prof. Mayank Baranwal, IIT Bombay
- **A First Course in Optimisation** | Prof. Debasattam Pal, IIT Bombay
- **Reinforcement Learning for Business Students** | Prof. Pekka Malo, Aalto University
- **Programming for Machine Learning** | e-PG Diploma in AI & Data Science, IIT Bombay
- **Makerspace, Introductory Robotics** and **Computer Programming and Utilisation** | IIT Bombay

Technical Skills

Languages & Libraries	Python, MATLAB, ROS, Linux, C, C++, C#
Softwares	PX4, MATLAB, Simulink, Gazebo, ArduinoIDE, Eagle, LaTeX

Key Courses

Robotics	Guidance and Control of UAVs, Control systems, Adaptive & Learning Control, Multivariable Control, Optimal Control, Computer Vision
Math & CS	Applied Linear Algebra, Linear Algebra Differential Equations, RL, Optimization, Nonlinear Systems, Intro to DS and ML, Distributed Optimization and ML
Electrical	Microprocessors, Digital Systems, Analog Circuits, Probability , Signal Processing